Build Your Own Rule-Based Python Chatbot

30-Minute Student Task — Completed Submission

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Task Overview

The goal of this task was to build a simple rule-based conversational chatbot in Python. The chatbot asks for the user's name, runs continuously inside a **while True** loop, cleans each input with **.lower().strip()**, recognizes multiple greetings and topics using **if / elif / else** logic, uses the **in** operator for flexible matching, and exits cleanly with **break** when the user types bye or exit.

Requirements Checklist

1. Asks the user's name before the main conversation
2. Uses while True: to keep the conversation running
3. Uses input() to receive messages
4. Cleans input using .lower().strip()
5. Recognizes multiple greetings (hello, hi) via the in operator
6. Answers multiple topics: name, how are you, time, weather, help
7. Uses if / elif / else for decision making
8. Uses break to exit on bye / exit
9. Uses the user's name in responses
10. Provides a fallback response for unknown messages via else

Program Workflow

The chatbot follows a continuous five-stage loop until the user chooses to exit:



The loop repeats after every RESPOND step, returning to INPUT, until the user types bye or exit and the program reaches EXIT via **break**.

Python Code — chatbot.py

chatbot.py

```
# Rule-Based Python Chatbot
# A simple conversational chatbot that greets the user, answers a few
# fixed topics, and exits when asked.

def main():
    print("Chatbot: Hello! Before we start, what is your name?")
    name = input("You: ").strip().title()

    print(f"Chatbot: Nice to meet you, {name}! Ask me something, or type 'bye' to exit.")

    while True:
        user_input = input("You: ").lower().strip()

        if user_input in ["bye", "exit"]:
            print(f"Chatbot: Goodbye, {name}! Have a great day.")
            break

        elif "hello" in user_input or "hi" in user_input:
            print(f"Chatbot: Hello there, {name}! How can I help you today?")

        elif "how are you" in user_input:
            print("Chatbot: I'm doing well, thanks for asking!")

        elif "your name" in user_input:
            print("Chatbot: I'm a simple rule-based chatbot built in Python.")

        elif "time" in user_input:
            print("Chatbot: Sorry, I can't check the time. Try your system clock!")

        elif "weather" in user_input:
            print("Chatbot: I can't check live weather, but I hope it's nice out!")

        elif "help" in user_input:
            print("Chatbot: Ask me my name, how I'm doing, the time, or weather.")

        else:
            print(f"Chatbot: Sorry {name}, I don't understand that. Try something else.")


if __name__ == "__main__":
    main()
```

Sample Test Run

The transcript below demonstrates the required test cases: hello, HELLO, hello, hi, a recognized question, an unrecognized message, and the exit command.

Console Output

```
Chatbot: Hello! Before we start, what is your name?
You: Sara
Chatbot: Nice to meet you, Sara! Ask me something, or type 'bye' to exit.

You: hello
Chatbot: Hello there, Sara! How can I help you today?

You: HELLO
Chatbot: Hello there, Sara! How can I help you today?

You: hello
Chatbot: Hello there, Sara! How can I help you today?

You: hi
Chatbot: Hello there, Sara! How can I help you today?

You: how are you
Chatbot: I'm doing well, thanks for asking!

You: what is your name
Chatbot: I'm a simple rule-based chatbot built in Python.

You: asdkjasjd
Chatbot: Sorry Sara, I don't understand that. Try something else.

You: bye
Chatbot: Goodbye, Sara! Have a great day.
```

All ten requirements have been met and the chatbot was verified against every listed test case.